Neural Nexus Submission

Team Leader Name: Jagdish Bhatt
Team Member(s) Name: Krishna Pandey , Saurav chuphal , Mayank kandpal

Team : Overclocked

Graphic Era Deemed to be University DEHRADUN

GRAFEST

Problem : **Lack of Real-time Classroom behavioural insights**

To develop and desgin a ai/ml based system which can analyse multimodal classroom behavior from the given dataset , it aims to detect and classify student engagement and activity during a live teaching session.

1. Teacher can’t monitor every student simultaneously .
2. No system to quantify , engagement , distraction & participation.
3. Existing system focus only on surveillance not on intelligence

Graphic Era Deemed to be University DEHRADUN

GRAFEST

Verify: 0086_000838.jpg

Our Solution :

AI Powered Classroom Intelligance system

1. Multimodal behaviour detection :- talking , read, handraise, etc
2. Identify tracking of students .
3. Temporal behavioural analysis .
4. Real-time engagement & disruption matrics

We convert Raw Classroom Video → Actionable video insights

Graphic Era Deemed to be University DEHRADUN

GRAFEST

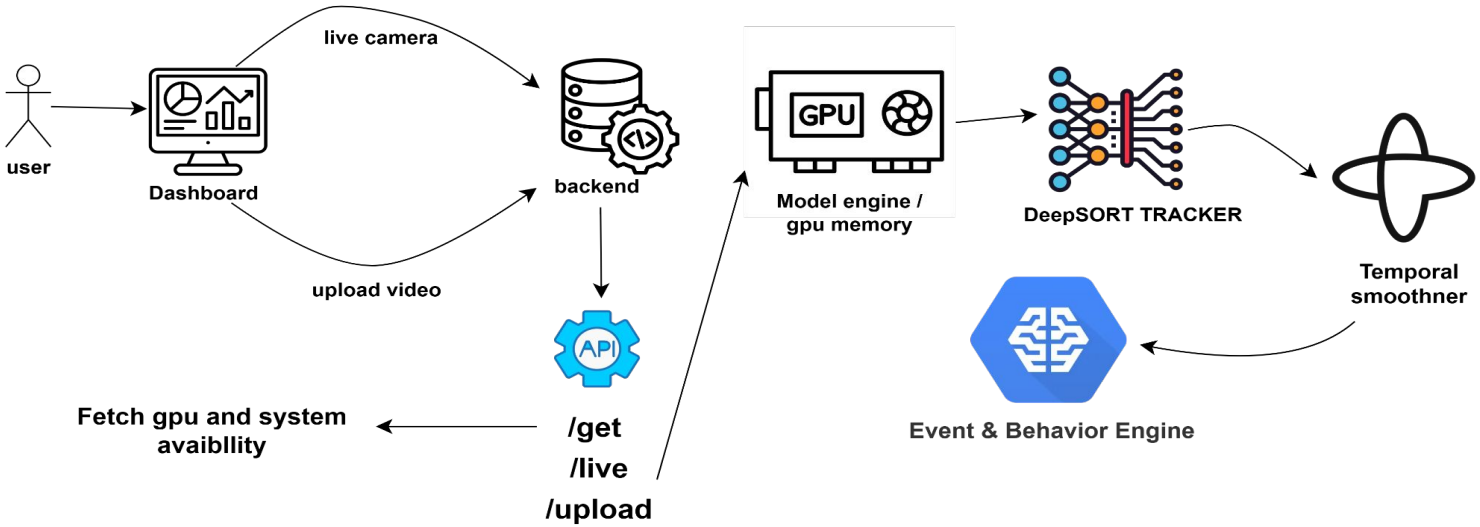
Techstack

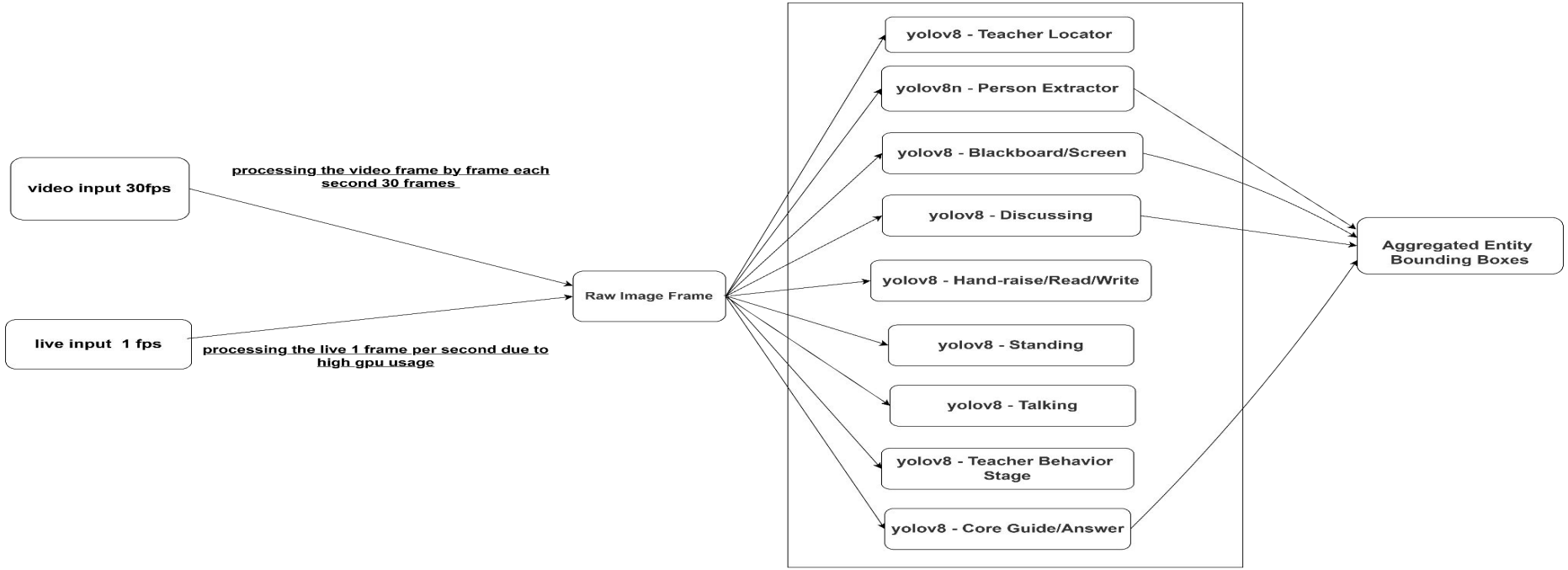
1. **Yolo** (FINE TUNED)
2. Deepsort
3. Temporal Logic (Sliding Window)

Backend :

1. Python
2. FastAPI

Architecture Diagram

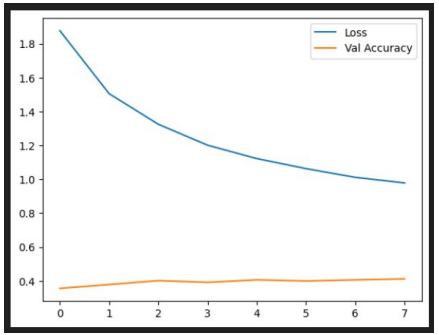


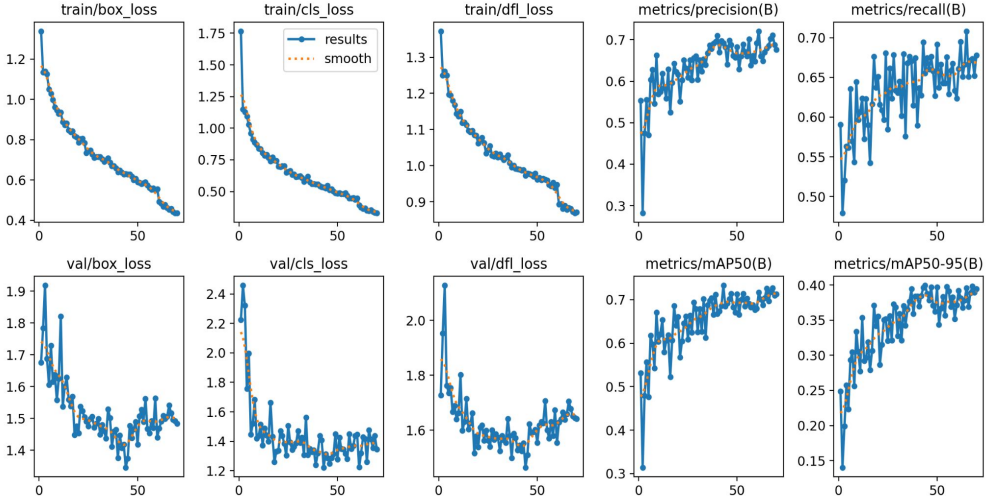

video input 30fps
processing the video frame by frame each second 30 frames
live input 1 fps
processing the live 1 frame per second due to high gpu usage
Raw Image Frame
yolov8 - Teacher Locator
yolov8n - Person Extractor
yolov8 - Blackboard/Screen
yolov8 - Discussing
yolov8 - Hand-raise/Read/Write
yolov8 - Standing
yolov8 - Talking
yolov8 - Teacher Behavior Stage
yolov8 - Core Guide/Answer
Aggregated Entity Bounding Boxes

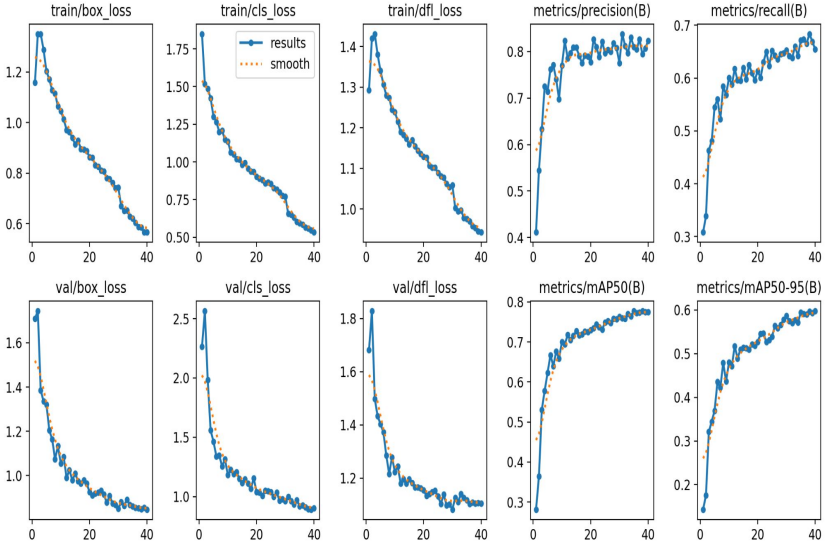
Graphic Era Deemed to be University DEHRADUN

GRAFEST

Evaluation







Graphic Era Deemed to be University DEHRADUN

GRAFEST

Prototype

Graphic Era Deemed to be University DEHRADUN

GRAFEST

Prototype